

NeuroFlow CFD Analyst

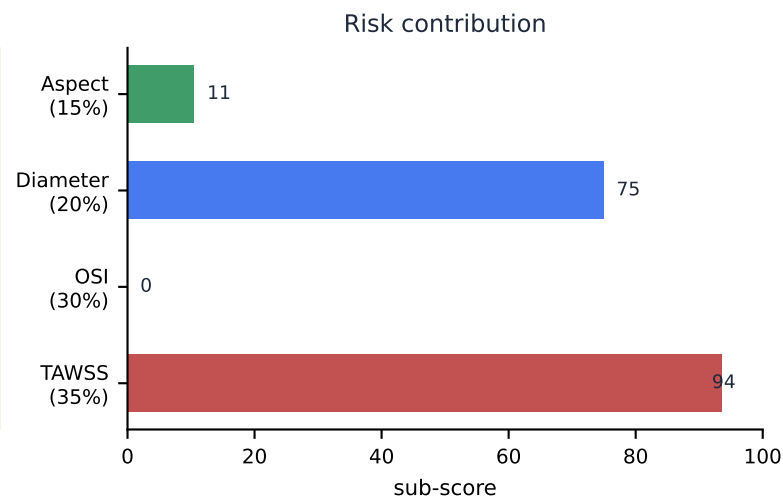
Clinical Hemodynamic Assessment

Case PT-2026-0101 · generated 2026-08-04 14:43

49

Composite Risk Index / 100

Moderate



Hemodynamic parameters

Parameter	Value	Threshold	Status
TAWSS — aneurysm dome	0.236 Pa	< 0.4 Pa	FLAGGED
TAWSS — parent artery	2.969 Pa	reference	normal
OSI — aneurysm dome	0.000	> 0.2	normal
RRT — aneurysm dome	11.04 Pa ⁻¹	> 3.0	FLAGGED
ECAP — aneurysm dome	0.000 Pa ⁻¹	> 1.0	normal
NWSS (sac / parent)	0.080	—	normal
LSAR (relative, <10% parent)	84.7 %	—	normal
LSAR (absolute, <0.4 Pa)	88.2 %	—	normal

Morphology (measured from the reconstructed surface)

Max diameter	Neck diameter	Aspect ratio	Dome-to-neck
8.00 mm	6.87 mm	0.89	1.16
Volume	Surface area	Non-sphericity	
241.4 mm ³	175.9 mm ²	0.154	

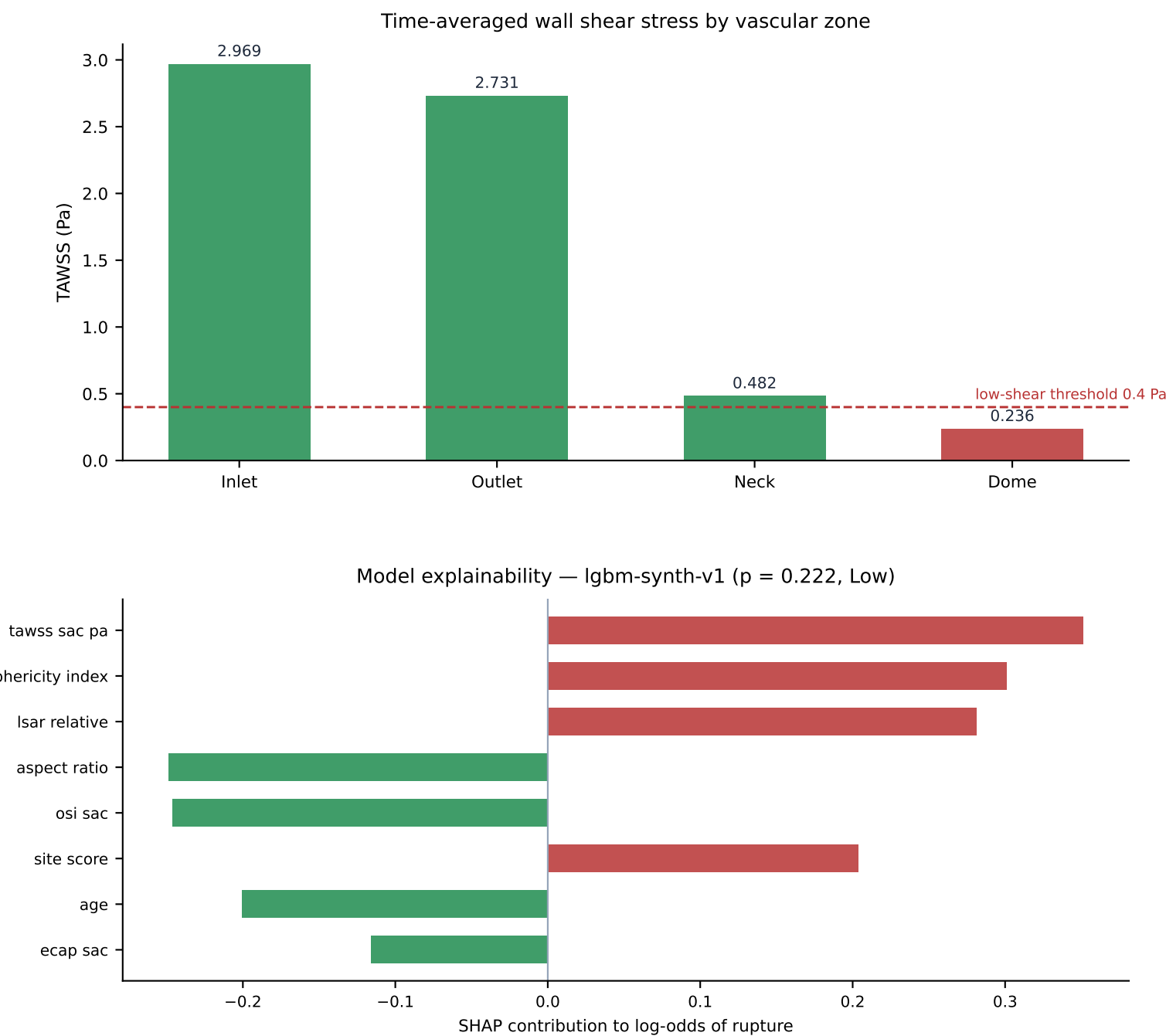
Provenance

Solver: OpenFOAM ESI v2412 · Mesh: cells: 522031 · SIMPLE solution converged in 336 iterations
WSS converted from OpenFOAM kinematic units (m^2/s^2) to Pascals by multiplying by rho = 1060.0. Validated against the analytic Poiseuille solution tau = 4*mu*Q/(pi*r^3).

NeuroFlow CFD Analyst

Zone Analysis and Model Explainability

Case PT-2026-0101



Assessment

Computational fluid dynamics analysis of case PT-2026-0101. Time-averaged wall shear stress at the aneurysm dome measured 0.236 Pa against 2.969 Pa in the parent artery (normalised WSS 0.080), with an oscillatory shear index of 0.000. 84.6% of the sac surface lies below 10% of the parent-artery shear (LSAR). Relative residence time at the dome is 11.04 Pa⁻¹. Sac morphology measured from the reconstructed surface: maximum diameter 8.00 mm, neck 6.87 mm, aspect ratio 0.89, volume 241.4 mm³. Composite Risk Index 49/100 (Moderate).